Fourier Transform

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\,dt \quad ; \quad x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft}\,dt$$

$c_1 x_1(t) + c_2 x_2(t)$	$\circ\!\!-\!\!\bullet^{F}\; c_1 X_1(f) + c_2 X_2(f)$
$x(t) * y(t)$	$\circ\!\!-\!\!\bullet\; X(f)\,Y(f)$
$x(t) \cdot y(t)$	$\circ\!\!-\!\!\bullet\; X(f) * Y(f)$
$x(t - t_0)$	$\circ\!\!-\!\!\bullet\; X(f) e^{-j2\pi f t_0}$
$e^{j2\pi f_0 t} x(t)$	$\circ\!\!-\!\!\bullet\; X(f - f_0)$
$x(at)$	$\circ\!\!-\!\!\bullet\; \frac{1}{\lvert a\rvert} X(f/a)$
$x(-t)$	$\circ\!\!-\!\!\bullet\; X(-f)$
$t \cdot x(t)$	$\circ\!\!-\!\!\bullet\; \frac{-1}{j2\pi} \frac{d}{df} X(f)$
$\frac{d}{dt} x(t)$	$\circ\!\!-\!\!\bullet\; j2\pi f\, X(f)$
$x^*(t)$	$\circ\!\!-\!\!\bullet\; X^*(-f)$
$X(t)$	$\circ\!\!-\!\!\bullet\; x(f)$
$\int_{-\infty}^{\infty} x^*(t) g(t)\,dt$	$\circ\!\!-\!\!\bullet\; \int_{-\infty}^{\infty} X^*(f) Y(f)\,df$
$\delta(t) \circ\!\!-\!\!\bullet 1$;	$1 \circ\!\!-\!\!\bullet \delta(f)$
$\text{Ш}_T(t)$	$\circ\!\!-\!\!\bullet\; \frac{1}{T} \text{Ш}_{1/T}(f)$
$\varepsilon(t)$	$\circ\!\!-\!\!\bullet\; \frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$
$\text{sgn}(t)$	$\circ\!\!-\!\!\bullet\; \frac{1}{j\pi f}$
$\frac{1}{\pi t}$	$\circ\!\!-\!\!\bullet\; -j\,\text{sgn}(f)$
$e^{-\lvert t\rvert/T}$	$\circ\!\!-\!\!\bullet\; \frac{2T}{1 + (2\pi f)^2}$
$\text{rect}(t/T)$	$\circ\!\!-\!\!\bullet\; \lvert T\rvert \,\text{si}(\pi T f)$
$\text{si}(\pi t/T)$	$\circ\!\!-\!\!\bullet\; \lvert T\rvert \,\text{rect}(Tf)$
$\Lambda(t/T)$	$\circ\!\!-\!\!\bullet\; \lvert T\rvert \,\text{si}^2(\pi T f)$
$\text{si}^2(\pi t/T)$	$\circ\!\!-\!\!\bullet\; \lvert T\rvert \,\Lambda(Tf)$
$\cos(2\pi f_0 t)$	$\circ\!\!-\!\!\bullet\; \frac{1}{2}[\delta(f+f_0) + \delta(f-f_0)]$
$\sin(2\pi f_0 t)$	$\circ\!\!-\!\!\bullet\; \frac{j}{2}[\delta(f+f_0) - \delta(f-f_0)]$
$e^{j2\pi f_0 t}$	$\circ\!\!-\!\!\bullet\; \delta(f - f_0)$
$r_{xy}(\tau)$	$\circ\!\!-\!\!\bullet^{E}\; X^*(f)\,Y(f)$

Problem with Power Signals

Magnitude & Phase response

$$H(f) = |H(f)|\, e^{j\varphi(f)}$$

$$|H(f)| = \sqrt{\text{Re}\{H(f)\}^2 + \text{Im}\{H(f)\}^2}$$

$$\varphi(f) = \arctan\left(\frac{\text{Im}\{H(f)\}}{\text{Re}\{H(f)\}}\right) = -b(f)$$

Phase Delay

$$t_p(f) = \frac{-\varphi(f)}{2\pi f}$$

Group Delay

$$t_g(f) = \frac{-1}{2\pi} \frac{d}{df} \varphi(f)$$

$$x(t) = x_{Re}(t) + x_{Ro}(t) + j\,x_{Ie}(t) + j\,x_{Io}(t)$$

$$X(f) = X_{Re}(f) + X_{Ro}(f) + jX_{Ie}(f) + jX_{Io}(f)$$

Fourier Series

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{j2\pi n f_0 t}\,dt \quad ; \quad X_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t) e^{-j2\pi f_0 t}\,dt$$

$$X(f) = \sum_{n=-\infty}^{\infty} X_n\, \delta(f - n f_0)$$

periodic time $\Rightarrow$ discrete spectrum

discrete time $\Leftarrow$ periodic spectrum

Fourier Transform Power Signals

$$X(f) = \lim_{\alpha \to 0} \int_{-\infty}^{\infty} x(t) e^{-\alpha|t|} e^{-j2\pi ft}\,dt$$

$$x(t) = \lim_{\alpha \to 0} \int_{-\infty}^{\infty} X(f) e^{-\alpha|t|} e^{j2\pi ft}\,dt$$

$$\sin^2(\alpha) = \frac{1}{2}(1 - \cos(2\alpha)) \qquad \int_0^T \cos/\sin = 0$$

$$\cos^2(\alpha) = \frac{1}{2}(1 + \cos(2\alpha))$$

$$\sin(\alpha)\cos(\beta) = \frac{1}{2}(\sin(\alpha - \beta) + \sin(\alpha + \beta))$$

$$\cos(\alpha)\cos(\beta) = \frac{1}{2}(\cos(\alpha - \beta) + \cos(\alpha + \beta))$$

$$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$$

$$\sin\alpha = \frac{1}{2}(e^{j\alpha} - e^{-j\alpha})$$

$$\cos\alpha = \frac{1}{2}(e^{j\alpha} + e^{-j\alpha})$$

$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$, with Konvergenz Region

$c_1 x_1(t) + c_2 x_2(t)$	$\circ\!\!-\!\!\bullet$	$c_1 X(s) + c_2 X(s)$, $K_1 \cap K_2$
$x(t) * y(t)$	$\circ\!\!-\!\!\bullet$	$X(s) Y(s)$, $K_x \cap K_y$
$x(t - t_0)$	$\circ\!\!-\!\!\bullet$	$e^{-s t_0} X(s)$, K_x
$e^{at} x(t)$	$\circ\!\!-\!\!\bullet$	$X(s-a)$, $K_x + \mathrm{Re}\{s\}$
$t x(t)$	$\circ\!\!-\!\!\bullet$	$\frac{-d}{ds} X(s)$, K_x
$\frac{d}{dt} x(t)$	$\circ\!\!-\!\!\bullet$	$s X(s)$, K_x
$\int_{-\infty}^{t} x(t) dt$	$\circ\!\!-\!\!\bullet$	$\frac{1}{s} X(s)$, $K_x \cap \{\mathrm{Re}\{s\} > 0\}$
$x(a t)$	$\circ\!\!-\!\!\bullet$	$\frac{1}{\lvert a \rvert} X(s/a)$, $a \cdot K_x$
$x^*(t)$	$\circ\!\!-\!\!\bullet$	$X^*(s^*)$, K_x
$\delta(t)$	$\circ\!\!-\!\!\bullet$	1, $s \in \mathbb{C}$
1	$\circ\!\!-\!\!\bullet$	doesn't exists
$\varepsilon(t)$	$\circ\!\!-\!\!\bullet$	$\frac{1}{s}$, $\mathrm{Re}\{s\} > 0$
$t \varepsilon(t)$	$\circ\!\!-\!\!\bullet$	$\frac{1}{s^2}$, $\mathrm{Re}\{s\} > 0$
$e^{at} \varepsilon(t)$	$\circ\!\!-\!\!\bullet$	$\frac{1}{s-a}$, $\mathrm{Re}\{s\} > \mathrm{Re}\{a\}$
$-e^{at} \varepsilon(-t)$	$\circ\!\!-\!\!\bullet$	$\frac{1}{s-a}$, $\mathrm{Re}\{s\} < \mathrm{Re}\{a\}$
$\frac{t^m}{m!} e^{at} \varepsilon(t)$	$\circ\!\!-\!\!\bullet$	$\frac{1}{(s-a)^{m+1}}$, $\mathrm{Re}\{s\} > \mathrm{Re}\{a\}$
$\sin(\omega_0 t) \varepsilon(t)$	$\circ\!\!-\!\!\bullet$	$\frac{\omega_0}{s^2 + \omega_0^2}$, $\mathrm{Re}\{s\} > 0$
$\cos(\omega_0 t) \varepsilon(t)$	$\circ\!\!-\!\!\bullet$	$\frac{s}{s^2 + \omega_0^2}$, $\mathrm{Re}\{s\} > 0$
$\mathrm{rect}(t/T)$	$\circ\!\!-\!\!\bullet$	$\frac{1}{s}(e^{sT/2} - e^{-sT/2})$, $s \in \mathbb{C}$

$X(f) = X(s)\big|_{j2\pi f}$ $\quad \omega_0 = 2\pi f_0$

$H(s) = \frac{b_M \prod_{k=1}^{M} (s - s_{0,k})}{b_N \prod_{k=1}^{N} (s - s_{\infty,k})} = \frac{Y(s)}{X(s)}$

$|H(s)|_{j2\pi f}| = |H(f)| = \left|\frac{b_M}{b_N}\right| \frac{\prod_{k=1}^{M} |j2\pi f - s_{0,k}|}{\prod_{l=1}^{N} |j2\pi f - s_{\infty,l}|}$

$\varphi(f) = \sphericalangle\left(\frac{b_M}{b_N}\right) + \sum_{k=1}^{M} \sphericalangle(j2\pi f - s_{0,k}) - \sum_{l=1}^{N} \sphericalangle(j2\pi f - s_{\infty,l})$

Allpass Filter: $s_{0,k} = -s_{\infty,k}^*$

Min. Phase Filter: → all zeros left

Max Phase Filter: → all zeros right!!

Stable: all poles left

Quasistable: all poles left & single pole on $\mathrm{Im}\{s\}$

Unstable: pole right & or multiple pole on $\mathrm{Im}\{s\}$

Causal System → Konvergenz right from the Pole!

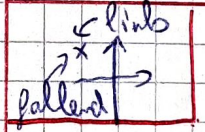
$|H(f)|$: poles → Extrema

at frequency $f = \frac{\mathrm{Im}(s_\infty)}{2\pi}$

Sampling without aliasing $\quad \cos(\omega t)$

$f_s \geq 2B$! $\quad f_s = \frac{1}{T_s}$, $B = \frac{\omega}{2\pi}$

$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\, \delta(t - nT_s)$



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DTFT

$\Omega = 2\pi f T_s$

$$X(\Omega) = \sum_{k=-\infty}^{\infty} x(k) e^{-j\Omega k}$$

$x(k-k_0) \circ\!-\!\bullet e^{-j\Omega k_0} X(\Omega)$ (DTFT)

$e^{j2\pi f_0 k} x(k) \circ\!-\!\bullet X(\Omega - \Omega_0)$

$x(k) y(k) \circ\!-\!\bullet \frac{1}{2\pi} X(\Omega) * Y(\Omega)$

$x(k) * y(k) \circ\!-\!\bullet X(\Omega) Y(\Omega)$

$x^*(k) \circ\!-\!\bullet X^*(-\Omega)$

$x^*(-k) \circ\!-\!\bullet X^*(\Omega)$

$k x(k) \circ\!-\!\bullet j \frac{d}{d\Omega} X(\Omega)$

$x(k) = x_{Re}(k) + x_{Ro}(k) + j x_{Ie}(k) + j x_{Io}(k)$

$X(f) = X_{Re}(f) + X_{Ro}(f) + jX_{Ie}(f) + jX_{Io}(f)$

Z-Transform

$$X(z) = \sum_{k=-\infty}^{\infty} x(k) z^{-k}$$

$x(k) * y(k) \circ\!-\!\bullet X(z) Y(z)$

$x(k-k_0) \circ\!-\!\bullet z^{-k_0} X(z)$

$a^k x(k) \circ\!-\!\bullet X(z/a)$

$k x(k) \circ\!-\!\bullet -z \frac{d}{dz} X(z)$

$x^*(k) \circ\!-\!\bullet X^*(z^*)$

$x(-k) \circ\!-\!\bullet X(\frac{1}{z})$, $1/k_x$

$a^k \varepsilon(k) \circ\!-\!\bullet \frac{z}{z-a}$, $|z| > |a|$

$-a^k \varepsilon(-k-1) \circ\!-\!\bullet \frac{z}{z-a}$, $|z| < |a|$

$a^{k-1} \varepsilon(k-1) \circ\!-\!\bullet \frac{1}{z-a}$, $|z| > |a|$

$-a^{k-1} \varepsilon(-k) \circ\!-\!\bullet \frac{1}{z-a}$, $|z| < |a|$

$$\sum_{k=0}^{N-1} e^{-j\frac{2\pi}{N} \cdot nk} = \begin{cases} N, & l \cdot N, l \in \mathbb{Z} \\ 0, & \text{else} \end{cases}$$

All pass Filter: $z_{0k} = \frac{1}{z_{0k}^*}$

Min Phase Filter: all zeros inside

Max Phase Filter: all zeros outside

DFT

$$X(n) = \sum_{k=0}^{N-1} x(k) e^{-j\frac{2\pi nk}{N}}$$

$$x(k) = \frac{1}{N} \sum_{n=0}^{N-1} X(n) e^{j\frac{2\pi kn}{N}}$$

$x(k-i)_{\text{mod } N} \bullet\!-\!\circ e^{-j2\pi ni/N} X(n)$ (DFT)

$e^{j2\pi ik/N} x(k) \circ\!-\!\bullet X(n-i)_{\text{mod } N}$

$\sum_{q=0}^{N-1} x_1(k-q)_{\text{mod } N} x_2(q) \circ\!-\!\bullet X_1(n) X_2(n)$

$x_1(k) x_2(k) \circ\!-\!\bullet \frac{1}{N} \sum_{q=0}^{N-1} X_1(n-q)_{\text{mod } N} X_2(n)$

even Signal: $X_e(k) = X_e(N-k)$

odd Signal: $X_o(k) = -X_o(N-k)$

$$a_0 \sum_{k=0}^{N} q^k = \begin{cases} a_0 \frac{1-q^{N+1}}{1-q}, & q \neq 1 \\ a_0 (N+1), & q = 1 \end{cases} \qquad a_0 \sum_{k=0}^{\infty} q^k = \frac{a_0}{1-q}, \; |q| < 1$$

$$X_\alpha(s) = \sum_{k=-\infty}^{\infty} x(k) e^{-sT_A k} \Rightarrow X(z) = \sum_{k=-\infty}^{\infty} x(k) z^{-k}$$

$z = e^{-sT_A}$

$\alpha = 0 \Downarrow$

$z = e^{j\Omega}$

$$X(\Omega) = \sum_{k=-\infty}^{\infty} x(k) e^{-j\Omega k}$$

Stable if unite circle is included in Region

causal systems → Convergence outside of a circle!

FIR-Filter: $f_0 \to \otimes \; \cdots \; f_M \to \otimes$: $y(k) = \sum_{\mu=0}^{M} f_\mu x(k-\mu)$

IIR-Filter: $f_0 \to \otimes \; \cdots \; f_M \to \otimes$, $b_1 \to \otimes \; \cdots \; b_N \to \otimes$: $y(k) = \sum_{\mu=0}^{M} f_\mu x(k-\mu) - \sum_{\nu=1}^{N} b_\nu y(k-\nu)$

Transferfunction:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{\mu=0}^{M} f_\mu z^{-\mu}}{\sum_{\nu=0}^{N} b_\nu z^{-\nu}} = z^{N-M} f_0 \frac{\prod_{\mu=0}^{M} (z - z_{0,\mu})}{\prod_{\nu=1}^{N} (z - z_{\infty,\nu})}$$

$$|H(\Omega)| = |f_0| \frac{\left| \prod_{\mu=0}^{M} (e^{j\Omega} - z_{0\mu}) \right|}{\left| \prod_{\nu=1}^{N} (e^{j\Omega} - z_{\infty\nu}) \right|} \qquad \varphi(\Omega) = \arg\{H(\Omega)\}$$

$$\sum_{k=-\infty}^{-1} a^k z^{-k} = 1 - \sum_{k=0}^{\infty} a^{-k} z^k$$

Cyclic Properties

$X(n) = X(n+iN)$

$x(k) = x(k+iN)$

Periodic Extension

$\tilde{x}(k) = \tilde{x}(k \bmod N)$

$= \tilde{x}(k)_{\text{mod } N}$

$x(t) \to \boxed{h(t)} \to y(t) = x(t) * h(t)$

$X(z) \to \boxed{H(z)} \to Y(z) = X(z) \cdot H(z)$

Power & Energy of Signals

continuous	discrete
Instantaneous Power	
$P_{x_i} = \lvert x(t)\rvert^2$	$P_{x_i} = \lvert x(k)\rvert^2$
Power of a signal	
$P_x = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \lvert x(t)\rvert^2 dt$	$P_x = \lim_{n\to\infty} \frac{1}{2n+1}\sum_{k=-n}^{n} \lvert x(k)\rvert^2$
Power of a periodics.	
$P_x = \frac{1}{T}\int_{t_0}^{t_0+T} \lvert x(t)\rvert^2 dt$	$P_x = \frac{1}{N}\sum_{k=k_0}^{k_0+N-1} \lvert x(k)\rvert^2 dH$
Energie of a signal	
$E_x = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2 dx$	$E_x = \sum_{k=-\infty}^{\infty} \lvert x(k)\rvert^2 dt$

Convolution

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} h_k(\tau)\, x(t-\tau)\, d\tau \qquad -(\tau - t)$$

$$y(t) = x(t) * \varepsilon(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$$

$$T = \frac{2\pi}{\omega}$$

$$x(t) \circledast y(t) = \int_{t_0}^{t_0+T} x(\tau)\, y(t-\tau)\, d\tau$$

$$x(t) * \delta(t-t_0) = x(t-t_0)$$

$$x_T(t) = x(t) * \sum_{k=-\infty}^{\infty} \delta(t-kT)$$

$$r_{xy}^{E}(\tau) = x^*(-\tau) * y(\tau)$$

$$r_{xy}^{P}(\tau) = \frac{1}{T}\, x(t) \circledast y(\tau)$$

Correlation

of Energy signals

$$r_{xy}^{E}(\tau) = \langle x(t), y(t+\tau)\rangle = \int_{-\infty}^{\infty} x^*(t)\, y(t+\tau)\, dt$$

$$r_{xy}^{E}(\tau) = \sum_{k=-\infty}^{\infty} x^*(k)\, y(k+\tau)$$

auto

$$r_{xx}^{E}(\tau) = \langle x(t), x(t)\rangle = \int_{-\infty}^{\infty} x^*(t)\, x(t+\tau)\, dt$$

$$E_x = r_{xx}^{E}(0)$$

of Power signals

cross

$$r_{xy}^{P}(\tau) = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} x^*(t)\, y(t+\tau)\, dt$$

$$r_{xy}^{P}(\tau) = \lim_{n\to\infty} \frac{1}{2n+1}\sum_{k=-n}^{n} x^*(k)\, y(k+\tau)$$

cross of periodic

$$r_{xy}^{P}(\tau) = \frac{1}{T}\int_{t_0}^{t_0+T} x^*(t)\, y(t+\tau)\, dt$$

$$P_x = r_{xx}^{P}(0) \qquad ; r_{xx}(\tau) = r_{xx}^*(-\tau)$$

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causal LTI system: $h(t) = 0,\ t < 0 \ \wedge\ \forall t > t_0$

BIBO: $\int_{-\infty}^{\infty} \lvert h(\tau)\rvert\, d\tau < \infty$

memoryless only $y(t_0)$ on $x(t_0)$

$x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

$$x_h(t) = \frac{1}{2}[x(t) + x^*(t)]$$

$$x_I(t) = \frac{1}{2i}[x(t) - x^*(t)]$$

conjugate even: $x(t) = x^*(-t)$

conjugate odd: $x(t) = -x^*(-t)$

$$\langle x(t), y(t)\rangle = \int_{-\infty}^{\infty} x^*(t)\, y(t)\, dt$$

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